

1 Summary

Model (Run-name): model3 (test)

Iteration: 15

TAC: 64127.02516267756

Took: 1770.662497997284s

Total Time: 6536.290684700012s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

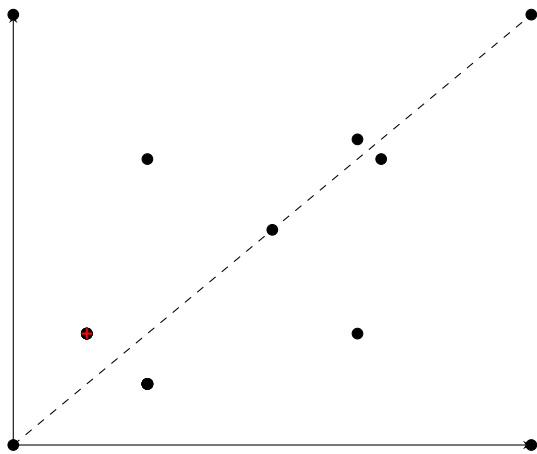
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{2,5,2}$	52.50497488801042	$q_{1,1,2}$	586.9600000000132
$z_{2,1,1}$	1.0	$dt_{2,5,3}$	44.5	$q_{2,1,1}$	173.03999999998678
$z_{2,2,1}$	1.0	$dt_{3,5,1}$	22.3	$q_{2,2,1}$	647.2099999999999
$z_{2,5,2}$	1.0	$dt_{3,5,2}$	27.492455765117473	$q_{2,5,2}$	357.41000000001304
$z_{3,5,1}$	1.0	$dt_{4,3,2}$	50.0	$q_{3,5,1}$	1186.8799999999767
$z_{4,3,2}$	1.0	$dt_{4,3,3}$	111.1	$q_{4,3,2}$	1539.72
$z_{5,4,2}$	1.0	$dt_{5,4,2}$	22.2	$q_{5,4,2}$	1634.85
$z_{cu,3}$	1.0	$dt_{5,4,3}$	24.335593220334786	$q_{cu,3}$	1197.4
$z_{cu,5}$	1.0	$dt_{cu,3}$	106.50540540540624	$q_{cu,5}$	724.5600000000002
$A_{1,1,2}^\beta$	6.886620958748166	$dt_{cu,5}$	66.535593220339	$\sigma_{1,1}^{(H)}$	160.0
$A_{2,1,1}^\beta$	2.132936236084747	$f_{1,1,2}^H$	8.8	$\sigma_{1,2}^{(H)}$	160.0
$A_{2,2,1}^\beta$	5.867260356733119	$f_{2,1,1}^H$	1.5575157515750389	$\sigma_{1,3}^{(H)}$	93.3
$A_{2,5,2}^\beta$	3.48131251943224	$f_{2,2,1}^H$	5.825472547254725	$\sigma_{2,1}^{(H)}$	248.9
$A_{3,5,1}^\beta$	10.353005046122822	$f_{2,5,2}^H$	10.6	$\sigma_{2,2}^{(H)}$	171.51792452830313
$A_{4,3,2}^\beta$	6.6758127195194445	$f_{3,5,1}^H$	7.367349472377252	$\sigma_{2,3}^{(H)}$	137.8
$A_{5,4,2}^\beta$	14.143774763563524	$f_{4,3,2}^H$	12.6	$\sigma_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	6.1446508890530085	$f_{5,4,2}^H$	13.188040829876567	$\sigma_{3,2}^{(H)}$	146.50540540540698
$A_{cu,5}^\beta$	5.324387898945314	$f_{1,1,2}^C$	5.86960000000013	$\sigma_{3,3}^{(H)}$	146.50540540540624
$A_{1,1,2}$	24.927399131526517	$f_{2,1,1}^C$	7.6	$\sigma_{4,1}^{(H)}$	271.1
$A_{2,1,1}$	3.551463972188699	$f_{2,2,1}^C$	6.1	$\sigma_{4,2}^{(H)}$	271.1
$A_{2,2,1}$	19.107703216652027	$f_{2,5,2}^C$	13.9	$\sigma_{4,3}^{(H)}$	148.9
$A_{2,5,2}$	8.599161402768438	$f_{3,5,1}^C$	13.9	$\sigma_{5,1}^{(H)}$	198.9
$A_{3,5,1}$	49.559658500720396	$f_{4,3,2}^C$	8.4	$\sigma_{5,2}^{(H)}$	198.9
$A_{4,3,2}$	23.668654733871367	$f_{5,4,2}^C$	17.3	$\sigma_{5,3}^{(H)}$	106.535593220339
$A_{5,4,2}$	82.71968770372072	$RecLMTD_{1,1,2}$	0.03609835297430287	$\sigma_{1,1}^{(C)}$	160.0
$A_{cu,3}$	20.613988155162765	$RecLMTD_{2,1,1}$	0.01744535585044224	$\sigma_{1,2}^{(C)}$	137.23157894737017
$A_{cu,5}$	16.234939065695293	$RecLMTD_{2,2,1}$	0.02509471073400322	$\sigma_{1,3}^{(C)}$	60.0
$dt_{1,1,2}$	22.768421052629833	$RecLMTD_{2,5,2}$	0.02045070700971767	$\sigma_{2,1}^{(C)}$	221.7
$dt_{1,1,3}$	33.3	$RecLMTD_{3,5,1}$	0.040271204826525964	$\sigma_{2,2}^{(C)}$	115.6
$dt_{2,1,1}$	88.9	$RecLMTD_{4,3,2}$	0.01306624355323738	$\sigma_{2,3}^{(C)}$	115.6
$dt_{2,1,2}$	34.286345580932974	$RecLMTD_{5,4,2}$	0.04300806468371278	$\sigma_{3,1}^{(C)}$	221.1
$dt_{2,2,1}$	27.2	$RecLMTD_{cu,3}$	0.014633280384072207	$\sigma_{3,2}^{(C)}$	221.1
$dt_{2,2,2}$	55.917924528303146	$RecLMTD_{cu,5}$	0.019045625215083236	$\sigma_{3,3}^{(C)}$	37.8

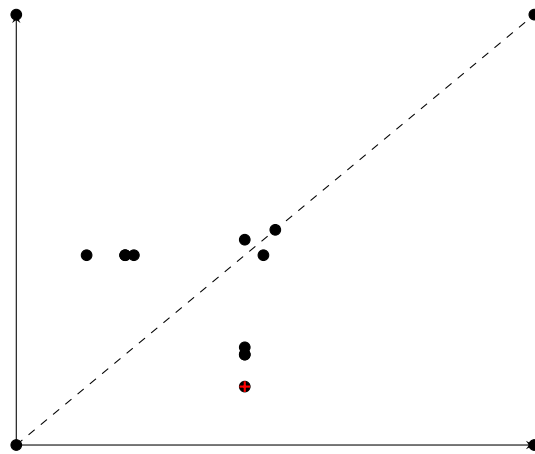
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,2}^{H,out}$	988.2513210624506
$t_{4,2}^{(C)}$	176.7	$b_{1,1,2}^{C,in}$	352.17600000000078
$t_{4,3}^{(C)}$	82.2	$b_{2,1,1}^{C,in}$	1042.96000000000132
$t_{5,1}^{(C)}$	204.4	$b_{2,2,1}^{C,in}$	705.16
$t_{5,2}^{(C)}$	119.01294964028946	$b_{2,5,2}^{C,in}$	1296.8699999999997
$t_{5,3}^{(C)}$	93.3	$b_{3,5,1}^{C,in}$	1654.28000000000284
$t_{1,1,2}^H$	93.3	$b_{4,3,2}^{C,in}$	317.52
$t_{2,1,1}^H$	138.25199893230317	$b_{5,4,2}^{C,in}$	1422.06000000000002
$t_{2,2,1}^H$	138.21791288913977	$b_{1,1,2}^{C,out}$	939.13600000000211
$t_{2,5,2}^H$	137.8	$b_{2,1,1}^{C,out}$	1216.0
$t_{3,5,1}^H$	65.6	$b_{2,2,1}^{C,out}$	1352.37
$t_{4,3,2}^H$	148.9	$b_{2,5,2}^{C,out}$	1654.28000000000127
$t_{5,4,2}^H$	72.55569732330778	$b_{3,5,1}^{C,out}$	2841.16000000000003
$t_{1,1,2}^C$	157.47461526361255	$b_{4,3,2}^{C,out}$	1857.24
$t_{2,1,1}^C$	160.0	$b_{5,4,2}^{C,out}$	3056.9099999999999
$t_{2,2,1}^C$	221.7		
$t_{2,5,2}^C$	119.0129496402887		
$t_{3,5,1}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,4,2}^C$	176.7		
$b_{1,1,2}^{H,in}$	1408.0		
$b_{2,1,1}^{H,in}$	387.6656705670272		
$b_{2,2,1}^{H,in}$	1449.960117011701		
$b_{2,5,2}^{H,in}$	1818.09000000000033		
$b_{3,5,1}^{H,in}$	1670.178125387923		
$b_{4,3,2}^{H,in}$	3415.86		
$b_{5,4,2}^{H,in}$	2623.1013210624496		
$b_{1,1,2}^{H,out}$	821.04000000000001		
$b_{2,1,1}^{H,out}$	214.6256705670404		
$b_{2,2,1}^{H,out}$	802.750117011701		
$b_{2,5,2}^{H,out}$	1460.68000000000012		
$b_{3,5,1}^{H,out}$	483.29812538794636		
$b_{4,3,2}^{H,out}$	1876.14		

3 *RecLMTD* Tangent Points

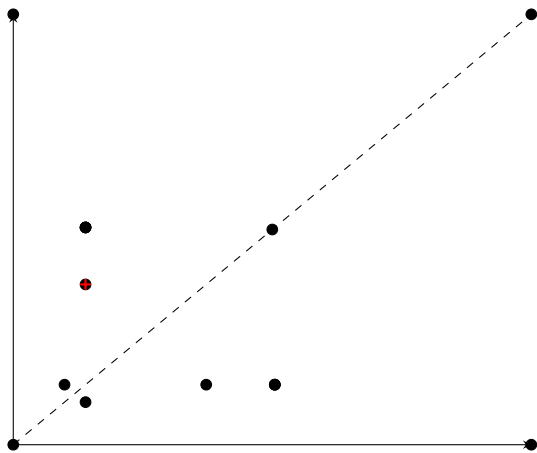
- (1, 1, 2)
 - Value: 0.036098
 - Error: 0.000000969314773
 - Relative error: 0.000026851328807
- (2, 1, 1)
 - Value: 0.017445
 - Error: 0.000000188963582
 - Relative error: 0.000010831624022
- (2, 2, 1)
 - Value: 0.025095
 - Error: 0.0000000000000003
 - Relative error: 0.0000000000000114
- (2, 5, 2)
 - Value: 0.020451
 - Error: 0.000213784512341
 - Relative error: 0.010345500740397
- (3, 5, 1)
 - Value: 0.040271
 - Error: 0.000042080875746
 - Relative error: 0.001043846340303
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000985435591
 - Relative error: 0.000075412743761
- (5, 4, 2)
 - Value: 0.043008
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000000
- $cu, 3$
 - Value: 0.014633
 - Error: 0.000000233609228
 - Relative error: 0.000015963987066
- $cu, 5$
 - Value: 0.019046
 - Error: 0.000000361898676
 - Relative error: 0.000019001308476



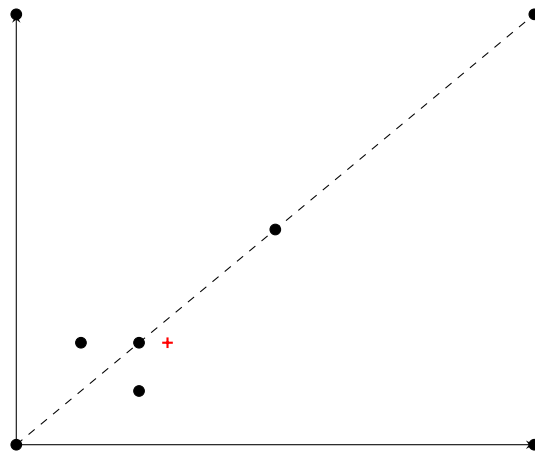
(a) RecLMTD - (1, 1, 2)



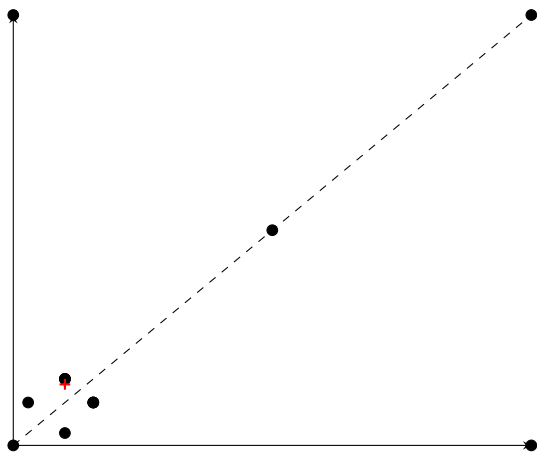
(b) RecLMTD - (2, 1, 1)



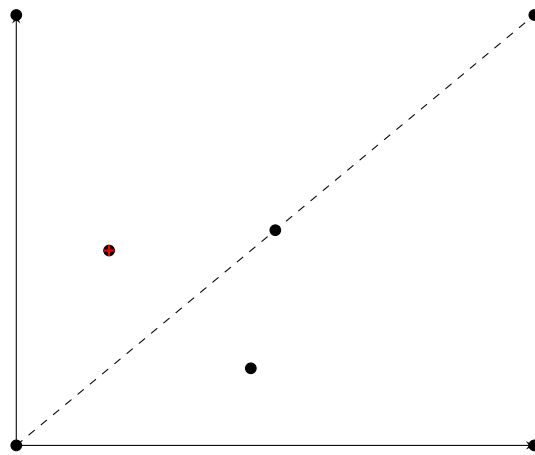
(c) RecLMTD - (2, 2, 1)



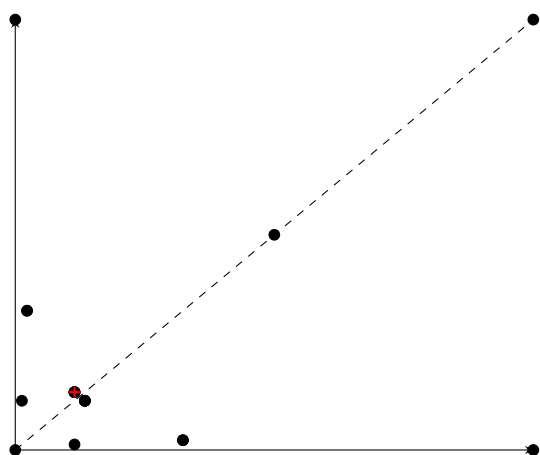
(d) RecLMTD - (2, 5, 2)



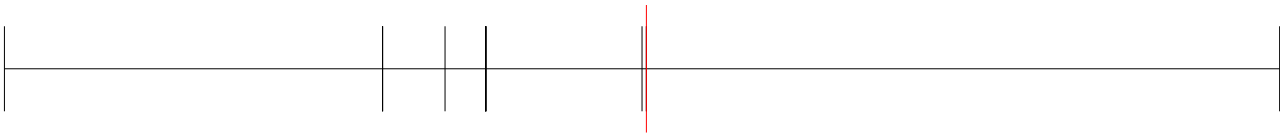
(e) RecLMTD - (3, 5, 1)



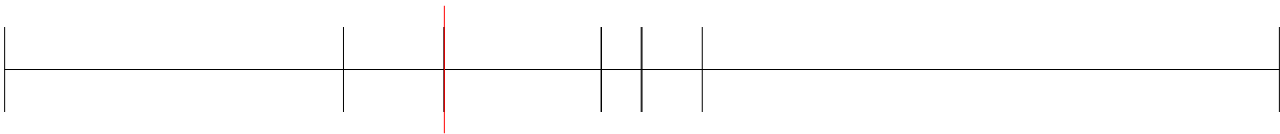
(f) RecLMTD - (4, 3, 2)



(g) *RecLMTD* - (5, 4, 2)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 387.6656705670272
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.960117011701
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 2)
 - Value: 1818.09000000000033
 - Absolute error: 9.777068044058979e-12
 - Relative error: -5.3776589960116984e-15
- (3, 5, 1)
 - Value: 1670.178125387923
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.361373807901006e-16
- (4, 3, 2)
 - Value: 3415.86
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 2623.1013210624496
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.733624802195118e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 821.04000000000001
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)

- Value: 214.6256705670404
 - Absolute error: 0.7039954567572408
 - Relative error: -0.003269384426943927
- (2, 2, 1)
 - Value: 802.750117011701
 - Absolute error: 2.4345400628276366
 - Relative error: -0.003023579798046615
- (2, 5, 2)
 - Value: 1460.6800000000012
 - Absolute error: 1.1368683772161603e-12
 - Relative error: 7.783144680670375e-16
- (3, 5, 1)
 - Value: 483.29812538794636
 - Absolute error: 1.3073986337985843e-12
 - Relative error: -2.7051597453417058e-15
- (4, 3, 2)
 - Value: 1876.14
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 988.2513210624506
 - Absolute error: 31.383822322501715
 - Relative error: 0.03279850383028946

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 352.17600000000078
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 1042.96000000000132
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0

- (2, 5, 2)
 - Value: 1296.8699999999997
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.7532495581147846e-16
- (3, 5, 1)
 - Value: 1654.2800000000284
 - Absolute error: 4.774847184307873e-12
 - Relative error: 2.886359736143703e-15
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 1422.0600000000002
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 2)
 - Value: 939.1360000000211
 - Absolute error: 14.8229982487004
 - Relative error: 0.01603677349622354
- (2, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 2)
 - Value: 1654.2800000000127
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.374457017211296e-16
- (3, 5, 1)
 - Value: 2841.1600000000003
 - Absolute error: 0.0

- Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.24
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 3056.9099999999999
 - Absolute error: 9.094947017729282e-13
 - Relative error: -2.9752092857589143e-16



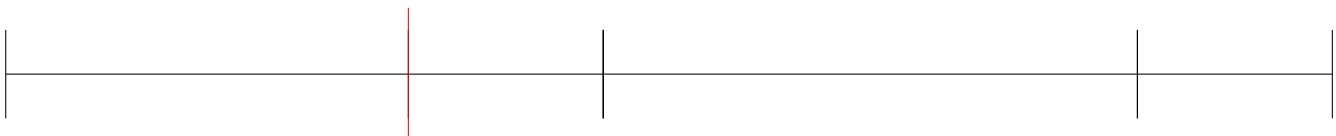
(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 1)$



(a) $t^{(H)} - (2, 1)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (3, 1)$



(a) $t^{(H)} - (4, 2)$



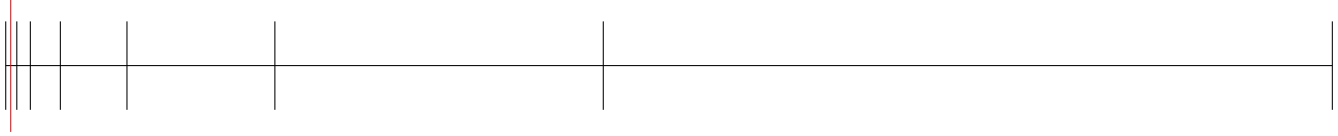
(a) $t^{(H)} - (5, 2)$



(a) $t^{(H)} - (1, 1, 2)$



(a) $t^{(H)} - (2, 1, 1)$



(a) $t^{(H)} - (2, 2, 1)$



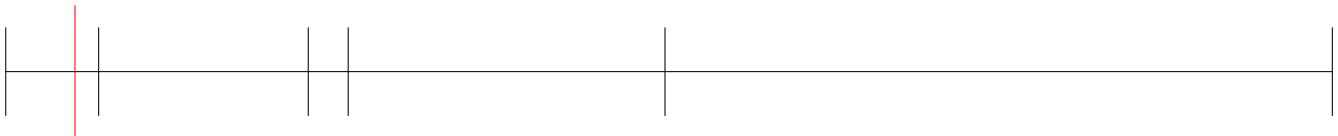
(a) $t^{(H)} - (2, 5, 2)$



(a) $t^{(H)} - (3, 5, 1)$



(a) $t^{(H)} - (4, 3, 2)$



(a) $t^{(H)} - (5, 4, 2)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (1, 2)$



$$(a) \ t^{(C)} - (2, 2)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (5, 2)$$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (4, 3)$$



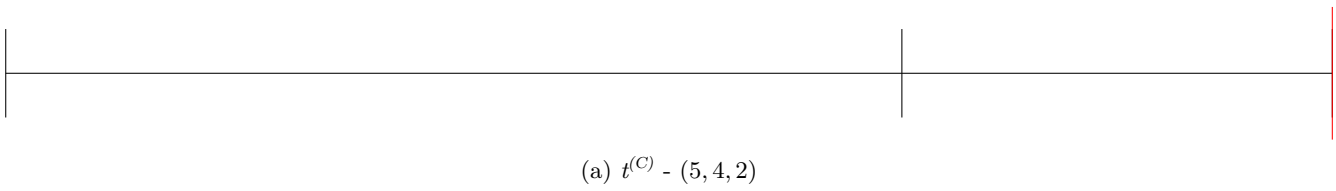
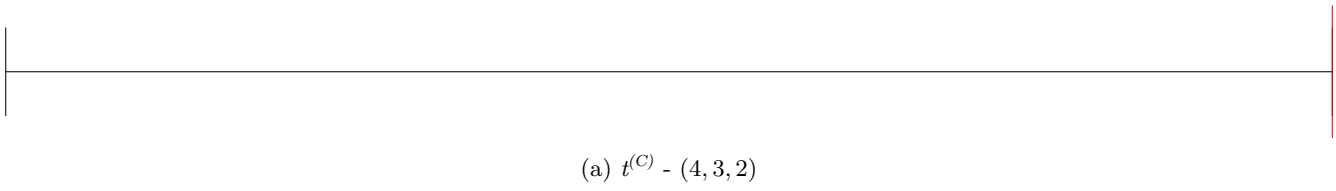
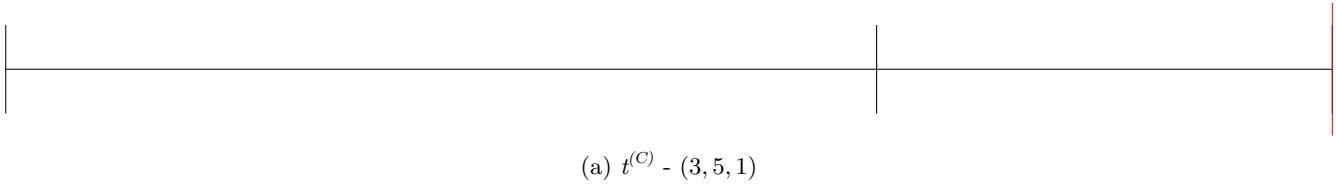
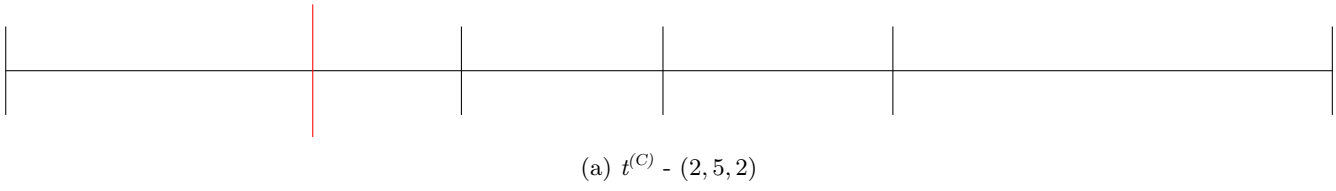
$$(a) \ t^{(C)} - (1, 1, 2)$$



$$(a) \ t^{(C)} - (2, 1, 1)$$



$$(a) \ t^{(C)} - (2, 2, 1)$$



5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 24.927399
 - Error: 0.000000000000295
 - Relative error: -0.000000000000012
- $(2, 1, 1)$
 - Value: 3.551464
 - Error: 0.000000000000117
 - Relative error: -0.000000000000033
- $(2, 2, 1)$
 - Value: 19.107703
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(2, 5, 2)$
 - Value: 8.599161
 - Error: 0.000000000011427
 - Relative error: -0.000000000001329
- $(3, 5, 1)$
 - Value: 49.559659
 - Error: 6.672209245757486
 - Relative error: 0.118655301933758
- $(4, 3, 2)$
 - Value: 23.668655
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(5, 4, 2)$
 - Value: 82.719688
 - Error: 0.000000000006139
 - Relative error: 0.000000000000074

5.2 A_{CU}

- $cu, 3$
 - Value: 20.613988
 - Error: 0.000000000000338
 - Relative error: -0.000000000000016
- $cu, 5$

- Value: 16.234939
- Error: 0.000000000000338
- Relative error: -0.000000000000021



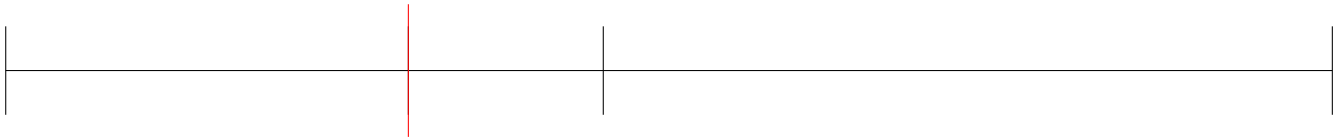
(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 1, 1)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (2, 5, 2)$



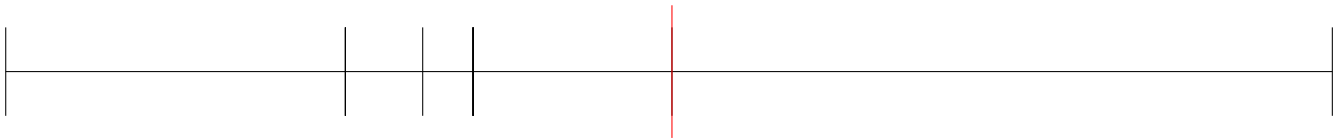
(a) $A^\beta - (3, 5, 1)$



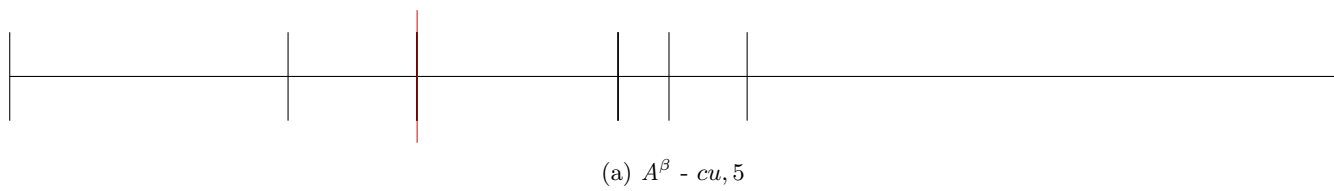
(a) $A^\beta - (4, 3, 2)$



(a) $A^\beta - (5, 4, 2)$



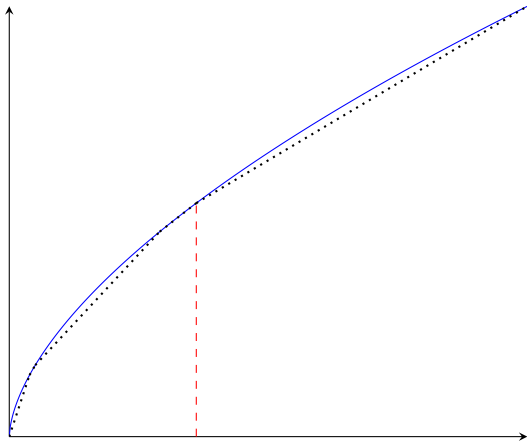
(a) $A^\beta - cu, 3$



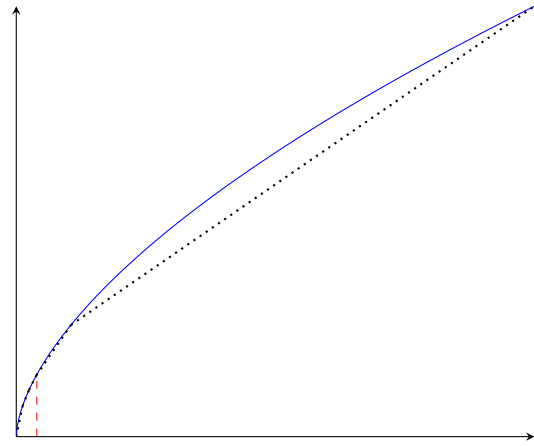
6 A^β Breakpoints

- (1, 1, 2)
 - q : 24.927399
 - $\hat{q}^\beta$: 6.886621
 - q^β : 6.886621
 - Breakpoints: [0, 2.7005882352941173, 3.460657649345137, 20.05363555217427, 24.927400386350968, 24.928068483287923, 24.928068483288385, 69.05411764705883]
 - Error: 0.000000008976277
 - Relative error: 0.000000001303437
- (2, 1, 1)
 - q : 3.551464
 - $\hat{q}^\beta$: 2.132936
 - q^β : 2.139166
 - Breakpoints: [0, 1.0776943916793027, 2.1978599029887658, 3.3036464983028657, 3.4652942623256235, 9.439495705511714, 89.41176470588236]
 - Error: 0.006229570448395
 - Relative error: 0.002912149413276
- (2, 2, 1)
 - q : 19.107703
 - $\hat{q}^\beta$: 5.867260
 - q^β : 5.871176
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 16.512573095835464, 16.51257309583564, 18.676744629514324, 18.890814020083003, 30.93278282491032, 30.932782824910646, 30.93278282491164, 76.14235294117645]
 - Error: 0.003915547099738
 - Relative error: 0.000666910200592
- (2, 5, 2)
 - q : 8.599161
 - $\hat{q}^\beta$: 3.481313
 - q^β : 3.636417
 - Breakpoints: [0, 2.702328746408619, 4.0106608195978595, 14.023793787177482, 138.54823529411763]
 - Error: 0.155104320970763
 - Relative error: 0.042653064205250
- (3, 5, 1)
 - q : 49.559659
 - $\hat{q}^\beta$: 10.353005
 - q^β : 10.401045
 - Breakpoints: [0, 30.750913778350093, 44.57376487924321, 71.16597394333012, 71.16597394333036, 71.16597394333074, 181.68117647058827]

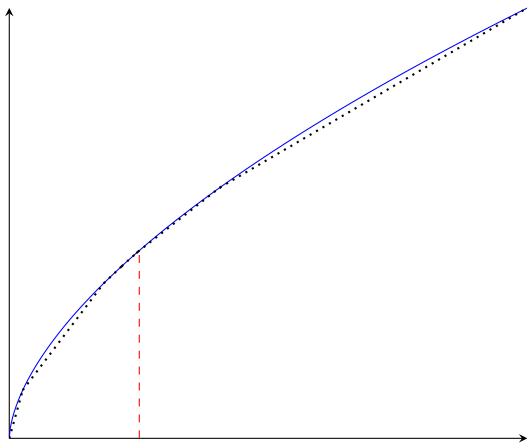
- Error: 0.048040139013411
- Relative error: 0.004618779955121
- (4, 3, 2)
 - q : 23.668655
 - $\hat{q}^\beta$: 6.675813
 - q^β : 6.675813
 - Breakpoints: [0, 19.92063386984238, 23.66863193022334, 23.668654733871367, 23.668654734065058, 23.668855770059167, 23.668855770059224, 181.14352941176475]
 - Error: 0.000000000019655
 - Relative error: 0.00000000002944
- (5, 4, 2)
 - q : 82.719688
 - $\hat{q}^\beta$: 14.143775
 - q^β : 14.143775
 - Breakpoints: [0, 11.133625686778567, 51.43476571725509, 51.6615457964989, 82.7196877037192, 82.71968770371944, 82.71968770371991, 106.27900213658211, 192.3352941176471]
 - Error: 0.000000000000092
 - Relative error: -0.000000000000007
- $cu, 3$
 - q : 20.613988
 - $\hat{q}^\beta$: 6.144651
 - q^β : 6.144651
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 13.643579101040302, 14.376341772690381, 16.738084386188994, 16.738084386191687, 16.73892021814917, 16.73899708924428, 16.738997089244634, 16.738997089244645, 18.852085704732907, 20.61398815516275, 128.44338870924773]
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $cu, 5$
 - q : 16.234939
 - $\hat{q}^\beta$: 5.324388
 - q^β : 5.324397
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.23492858720698, 16.234975184591587, 16.235135805353295, 20.944873037188618, 20.944873037188774, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.000008705031942
 - Relative error: 0.000001634933043



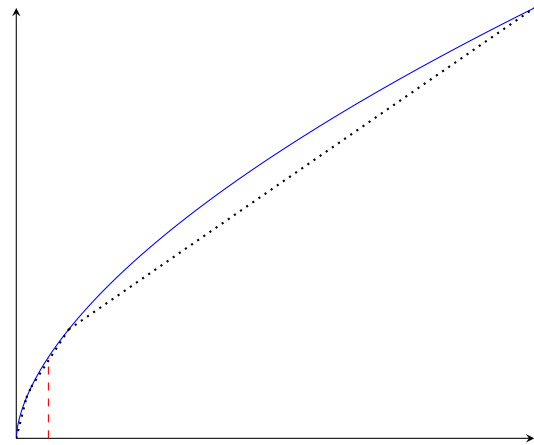
(a) $q^\beta - (1, 1, 2)$



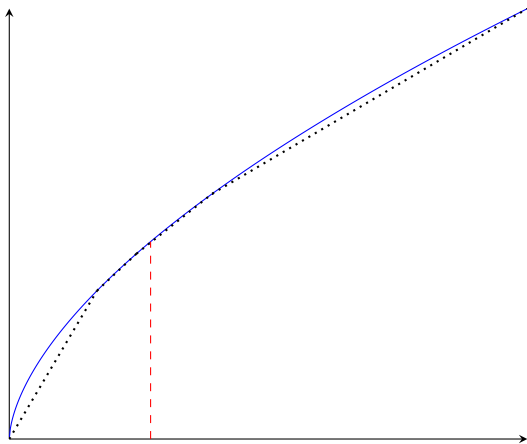
(b) $q^\beta - (2, 1, 1)$



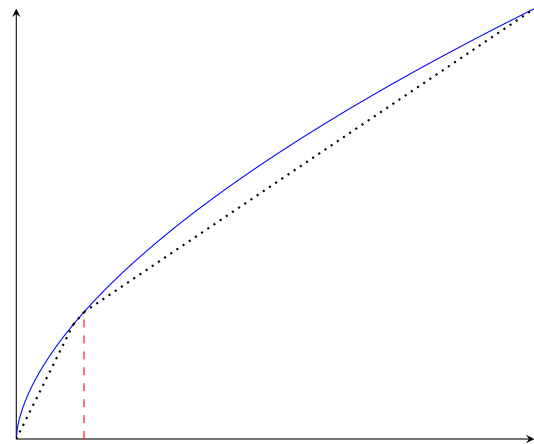
(c) $q^\beta - (2, 2, 1)$



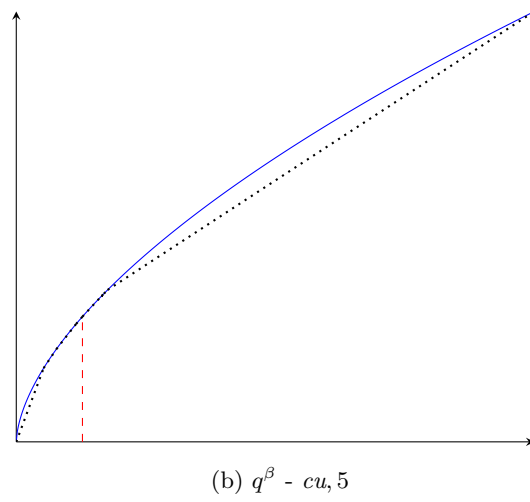
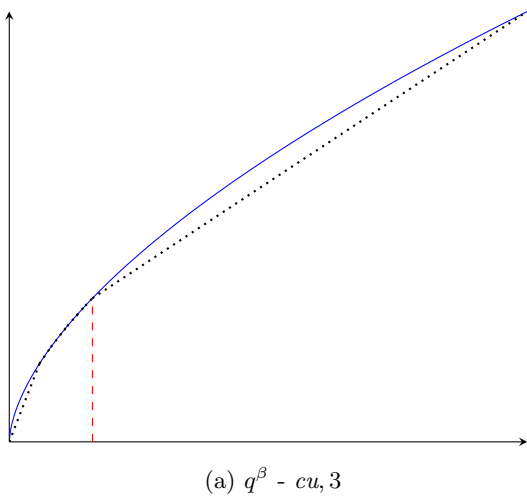
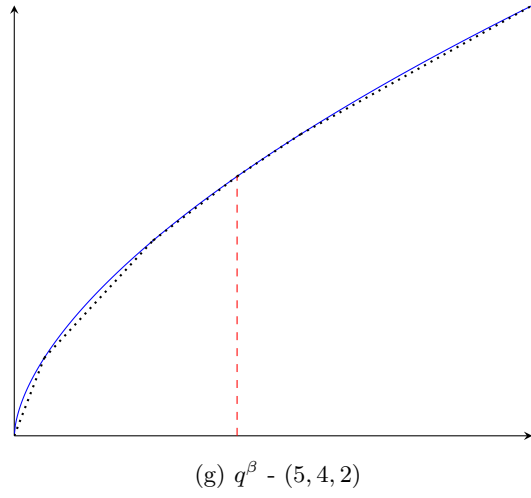
(d) $q^\beta - (2, 5, 2)$



(e) $q^\beta - (3, 5, 1)$



(f) $q^\beta - (4, 3, 2)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,2}$	-0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,1}$	-0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,1,2}$	-0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,2,2}$	-0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,3,2}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,1}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{5,5,2}$	-0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,2,2}$	-0.0	$z_{cu,1}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,1}$	-0.0	$z_{cu,2}$	-0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{cu,4}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,1}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,4,2}$	0.0	$z_{hu,2}$	-0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	-0.0	$z_{hu,4}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,2}$	0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,3,1}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,5,1}$	-0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,3}$	100.0
$A_{hu,5}^\beta$	0.0	$A_{4,5,1}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,1,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,1}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,2}$	33.717924528304565
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,4,1}$	0.0	$dt_{3,1,1}$	161.1
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,1,2}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,2,2}$	0.0	$A_{cu,1}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,1}$	0.0	$A_{cu,2}$	0.0	$dt_{3,2,2}$	111.1
$A_{2,3,2}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,3}$	10.0
$A_{2,4,1}$	0.0	$A_{hu,1}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,4,2}$	0.0	$A_{hu,2}$	0.0	$dt_{3,3,2}$	10.0
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,3}$	27.8
$A_{3,1,1}$	0.0	$A_{hu,4}$	0.0	$dt_{3,4,1}$	144.5
$A_{3,1,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,4,2}$	10.0
$A_{3,2,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,3}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,5,3}$	133.4
$A_{3,3,1}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{4,1,1}$	122.2
$A_{3,3,2}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{4,1,2}$	144.96842105262988
$A_{3,4,1}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	100.0
$A_{3,4,2}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,2,1}$	122.2
$A_{3,5,2}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,2,2}$	10.0
$A_{4,1,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	106.1
$A_{4,1,2}$	0.0	$dt_{1,4,2}$	10.0	$dt_{4,3,1}$	122.2
$A_{4,2,1}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,4,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,4,2}$	15.966949622056063
$A_{4,3,1}$	0.0	$dt_{1,5,2}$	10.0	$dt_{4,4,3}$	94.5

Variable	Value	Variable	Value	Variable	Value
$dt_{4,5,1}$	122.2	$f_{2,1,2}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,5,2}$	177.8	$f_{2,2,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,3,1}^H$	3.217011701170237	$f_{1,2,1}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,3,2}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,2}$	10.0	$f_{2,4,1}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	10.0	$f_{2,4,2}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,1}^H$	0.0	$f_{1,4,1}^C$	17.3
$dt_{5,2,2}$	74.66004781348396	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	10.0	$f_{3,1,2}^H$	7.583246480772167	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	7.432650527622749	$f_{1,5,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,2}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,4,1}$	116.7	$f_{3,3,2}^H$	7.216753519227834	$f_{2,3,1}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,2}$	10.0	$f_{3,4,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,1}$	53.3	$f_{4,1,1}^H$	12.6	$f_{2,5,1}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,2}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,3,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	8.4
$dt_{hu,4}$	63.3	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,1,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,2}^C$	1.730399999999868
$f_{1,3,2}^H$	0.0	$f_{5,2,2}^H$	4.511959170123427	$f_{4,2,1}^C$	0.0
$f_{1,4,1}^H$	8.8	$f_{5,3,1}^H$	0.0	$f_{4,2,2}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,3,1}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,4,1}^H$	0.0	$f_{4,4,1}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	17.7	$f_{4,4,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.0	$d_{1,3,2}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.0	$d_{1,4,1}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,5,2}$	0.0	$d_{1,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.011507146203332833	$d_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.01241539636733949	$d_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.0	$d_{2,1,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$d_{2,2,2}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.013066243553237336	$d_{2,3,1}$	0.0
$f_{5,4,1}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.017789138073956517	$d_{2,3,2}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.0195128108637384	$d_{2,4,1}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,5,1}$	0.0	$d_{2,4,2}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.0066792480523736215	$d_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.0	$d_{3,1,1}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.0	$d_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.0	$d_{3,2,1}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.0	$d_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.0	$d_{3,3,1}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028822002049121505	$d_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,1}$	0.014265344121837078	$d_{3,4,1}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.0	$d_{3,4,2}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.0	$d_{3,5,2}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.1	$\text{RecLMTD}_{cu,1}$	0.01610868052385036	$d_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115768875	$d_{4,1,2}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$d_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$d_{4,2,2}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.01607543104911131	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$d_{4,3,1}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$d_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$d_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$d_{4,5,1}$	0.0
$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,1,1}$	0.0	$d_{4,5,2}$	0.0
$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,1}$	0.0	$d_{5,1,1}$	0.0
$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,2,2}$	0.0	$d_{5,1,2}$	0.0
$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,1}$	0.0	$d_{5,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,1,2}^C$	60.0
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	69.95145202637727	$t_{2,3,1}^C$	221.1
$q_{5,4,1}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	65.6	$t_{2,4,1}^C$	97.2171088291546
$q_{5,5,2}$	0.0	$t_{3,5,2}^H$	65.6	$t_{2,4,2}^C$	82.2
$q_{cu,1}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,1}^C$	139.21155520203027
$q_{cu,2}$	0.0	$t_{4,1,2}^H$	271.1	$t_{3,1,1}^C$	127.37876463648115
$q_{cu,4}$	0.0	$t_{4,2,1}^H$	204.06666666666666	$t_{3,1,2}^C$	90.20394736842073
$q_{hu,1}$	0.0	$t_{4,2,2}^H$	271.1	$t_{3,2,1}^C$	115.6
$q_{hu,2}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	221.1
$q_{hu,4}$	0.0	$t_{4,4,2}^H$	271.1	$t_{3,3,2}^C$	37.8
$q_{hu,5}$	0.0	$t_{4,5,1}^H$	271.1	$t_{3,4,1}^C$	82.2
$t_{1,1,1}^H$	160.0	$t_{4,5,2}^H$	271.1	$t_{3,4,2}^C$	82.2
$t_{1,2,1}^H$	160.0	$t_{5,1,1}^H$	189.12372881355932	$t_{3,5,2}^C$	93.3
$t_{1,2,2}^H$	160.0	$t_{5,1,2}^H$	65.6	$t_{4,1,1}^C$	60.00012385978209
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,2}^C$	60.0
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	161.36528150681187	$t_{4,2,1}^C$	221.7
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,2}^C$	115.6
$t_{1,4,2}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,3,1}^C$	221.1
$t_{1,5,1}^H$	160.0	$t_{5,4,1}^H$	65.6	$t_{4,4,1}^C$	82.2
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,2}^C$	82.2
$t_{2,1,2}^H$	233.61200238186518	$t_{5,5,2}^H$	65.6	$t_{4,5,1}^C$	173.69194751441805
$t_{2,2,2}^H$	187.8424528301887	$t_{1,1,1}^C$	90.20407122820313	$t_{4,5,2}^C$	93.3
$t_{2,3,1}^H$	248.9	$t_{1,2,1}^C$	115.6	$t_{5,1,1}^C$	92.62148308308305
$t_{2,3,2}^H$	248.9	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	60.0
$t_{2,4,1}^H$	248.9	$t_{1,3,1}^C$	221.1	$t_{5,2,1}^C$	115.6
$t_{2,4,2}^H$	137.8	$t_{1,3,2}^C$	107.67619047619047	$t_{5,2,2}^C$	115.6
$t_{2,5,1}^H$	248.9	$t_{1,4,1}^C$	176.7	$t_{5,3,1}^C$	221.1
$t_{3,1,1}^H$	105.65126083492889	$t_{1,4,2}^C$	82.2	$t_{5,3,2}^C$	37.8
$t_{3,1,2}^H$	169.82926742494027	$t_{1,5,1}^C$	148.8544954686528	$t_{5,4,1}^C$	172.9875776163961
$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	148.8544954686528

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	124.01079136690645	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	897.4286789375498	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	1408.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,1}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	3520.5300000000007	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	800.714212421272	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	897.4286789375498
$b_{2,5,1}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	1408.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	1630.4594791482984	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,1}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	1684.981874612077	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	3520.5300000000007
$b_{3,2,2}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	537.8205208517293	$b_{2,3,1}^{H,out}$	800.714212421272	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.86	$b_{2,5,1}^{H,out}$	0.0	$b_{1,4,1}^{C,in}$	3056.9099999999994
$b_{4,1,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0	$b_{1,4,2}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	1630.4594791482984	$b_{1,5,1}^{C,in}$	0.0
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	1684.981874612077	$b_{1,5,2}^{C,in}$	0.0
$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0	$b_{2,1,2}^{C,in}$	0.0
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	705.16
$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	537.8205208517293	$b_{2,3,1}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	3056.9099999999994	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,1}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	1857.24	$b_{2,2,2}^{C,out}$	705.16		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	103.82399999999208	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	1857.24		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	103.82399999999208		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0		
$b_{5,4,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,5,1}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0		
$b_{1,2,1}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0		